

Guantian Zheng

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EDUCATION

Nanyang Technological University (NTU) 08/2026-06/2029

- Ph.D. in Robotics and Intelligent Systems, School of Mechanical and Aerospace Engineering
- AutomanLab Supervised by *Prof. Chen Lv*

Huazhong University of Science and Technology (HUST) 09/2022-06/2026

- Bachelor of Engineering in Integrated Circuit Design and Integrated System

RESEARCH INTERESTS

Autonomous Driving (World Model & End-to-End), Embodied AI, Computer Vision

PUBLICATIONS

*Equal contribution.

1. **Zheng, G.***, Xu, H., Gao, T., 2026.

Hyper²: Unleashing Hyperbolic Geometry's Full Potential through Dual-Space Consistency. *British Machine Vision Conference (BMVC 2026)*. **Accepted**.

2. **Zheng, G.***, Lyu, C., Yu, L., 2026.

YOUDrive: Driving Style as a Composable Direction for End-to-End Autonomous Driving. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2027)*. **Submitted**.

3. **Zheng, G.**, 2026.

StyleShield: Exposing the Fragility of AIGC Detectors through Continuous Controllable Style Transfer. *Annual Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026)*. **Under review**.

4. Ma, E.* , **Zheng, G.***, Zhang, J.* , Tang, T., Lu, Y., Yu, K., 2026.

DriveCombo: Benchmarking Compositional Traffic Rule Reasoning in Autonomous Driving. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026)*. **Accepted**.

5. Zhang, Z.* , **Zheng, G.***, Zou, S.* , Zhu, Z., Chi, G., Jiang, A., Zhao, H., 2026.

Unified Map Prior Encoder for Mapping and Planning. *IEEE International Conference on Robotics and Automation (ICRA 2026, Oral)*. **Accepted**.

6. Zhang, Z.* , **Zheng, G.***, Qiu, X.* , Zhang, B., Gilitschenski, I., Gu, X., Zhao, H., Zhao, H., 2025.

Delving into Mapping Uncertainty for Mapless Trajectory Prediction. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025, Oral)*. **Accepted**.

7. Zhang, Z., Li, X., Zou, S., **Zheng, G.**, Chi, G., Li, S., Zhao, H., Zhao, H., 2025.

Chameleon: Fast-slow Neuro-symbolic Lane Topology Extraction. *IEEE International Conference on Robotics and Automation (ICRA 2025, Oral)*. **Accepted**.

RESEARCH EXPERIENCES

Research Intern (World Model & End-to-End), Neolix 02/2026-Present

Team Leader, YouDrive: Driving Style as a Steerable Axis for Personalized End-to-End Driving (plan to submit to CVPR 2027)

- Proposed **YouDrive**, a personalized end-to-end planner on a **vision-language-action (VLA)** backbone that turns driving style into a steerable control axis
- Replaced the discrete action codebook with a **flow-matching decoder** that samples several dynamically feasible trajectories in a kinematic control space and commits to one
- Modeled each persona as a **low-rank weight edit (task vector)** in the VLA content space, so a single coefficient scales and composes styles without retraining the backbone
- Introduced the **Style Alignment Score (SAS)**; on NAVSIM navtest a single coefficient traces a monotone, controllable style path while keeping closed-loop safety above **0.90 PDMS** (up to 0.954)

Team Member, ATLAS: Large-Scale Multimodal Autonomous-Driving Backbone Pre-training

- Led pre-training of a large-scale multimodal autonomous-driving vision backbone, **ATLAS** (~7.5B parameters), under a **multi-teacher distillation** framework
- Combined **DINOv2/VGGT** (geometry & semantics), **Cosmos Tokenizer** (visual-generation supervision), and a **7B VLM** (semantic alignment) to jointly train a unified visual backbone
- Owned the **Video Head / Render Decoder**, realizing visual-token distillation via the Cosmos CI Tokenizer and systematically analyzing token-space alignment

Research Assistant, Autonomous Intelligence Lab (AutoLab), Westlake University 06/2025-01/2026

Advisor: Prof. Kaicheng Yu

Team Leader, **DriveCombo: Benchmarking Compositional Traffic Rule Reasoning in Autonomous Driving** (CVPR 2026)

- Built a **five-level cognitive-ladder** evaluation framework (**70K QA over 280K images**) to probe compositional traffic-rule reasoning of MLLMs beyond static perception benchmarks
- Developed a **Rule2Scene Agent** that automatically generates controllable **CARLA** scenarios from traffic rules for systematic edge-case and rule-conflict evaluation
- Benchmarked **14 MLLMs**; via targeted fine-tuning improved **rule-reasoning accuracy by 29.7%** and reduced **planning L2 error by 17.3%**

Research Assistant, Institute for AI Industry Research (AIR), Tsinghua University 06/2024-11/2025

Advisor: Prof. Hao Zhao

Team Leader, **Unified Map Prior Encoder for Mapping and Planning**

- Developed a **unified dual-branch encoder**: vector priors (**HD/SD maps**) via confidence-biased cross-attention, raster priors (**satellite, rasterized SD**) via FiLM-conditioned ResNet-18, enabling seamless fusion of arbitrary prior subsets
- Designed frame-wise SE(2) pre-alignment and multi-frequency sinusoidal point encoding for vector maps, with SE(2) micro-alignment for raster inputs, handling pose drift and scale mismatches
- Achieved **mAP +5.9 on MapTRv2** (61.5→67.4) and **+5.3 on MapQR** (66.4→71.7) across nuScenes/Argoverse2, and on end-to-end planning cut **trajectory L2 by 42%** (0.72→0.42m) and **collision rate by 45%**

Team Leader, **Delving into Mapping Uncertainty for Mapless Trajectory Prediction**

- Proposed a lightweight, self-supervised **Proprioceptive Scenario Gating** module that adaptively integrates map uncertainty into trajectory prediction based on the ego vehicle's future motion dynamics
- Designed a novel **Covariance-Based Map Uncertainty** model using 2D Gaussians to better capture road geometry, improving robustness over prior Laplace-based approaches
- Benchmarked four SOTA online map constructors (MapTR, MapTRv2, MapTRv2-Centerline, StreamMapNet) with HiVT/DenseTNT, achieving up to **23.6% gain** (minADE, minFDE, MR) and **10-30x** inference speedup

Team Member, **Chameleon: Fast-slow Neuro-symbolic Lane Topology Extraction**

- Developed a neuro-symbolic algorithm combining symbolic reasoning with **Chain-of-Thought VLMs**, and a program-synthesis framework generating executable Python from few-shot prompts, expert rules, and API descriptions
- Improved 3-shot lane-topology inference on **OpenLane-V2**, matching/outperforming supervised baselines while cutting inference from **>200s to 0.1-8s per frame**

Team Member, **OpenLane-V3: Multimodal Benchmark for Lane and Traffic-Element Reasoning**

- Extended OpenLane-V2 with additional **3D traffic-light and traffic-sign annotations** (semantic + positional) to evaluate multimodal cues for autonomous-driving perception and planning

INDEPENDENT PROJECTS

StyleShield: Continuous Controllable Style Transfer against AIGC Detectors

- Proposed a **flow-matching** framework for continuous, controllable text style transfer to systematically probe AIGC-detector robustness; a single parameter γ gives smooth strength control
- At $\gamma = 7.0$, achieved **94.6% evasion** on the trained detector and **$\geq 99\%$** on three unseen detectors while keeping **0.928 semantic similarity**; designed a RateAudit algorithm for arbitrary document-level detection-rate control

Hyper²: Dual-Space Consistency for Hyperbolic Point Cloud Completion

- Applied hyperbolic geometry to **both** feature embedding and loss, resolving HyperbolicCD's geometric inconsistency (feature-loss correlation **0.68→0.95**, gradient utilization **39%→87%**)
- Verified on ShapeNet-55/34, PCN, and KITTI with **+22.9%** performance and **+37.5%** unseen-category generalization at only 1.6% FLOPs overhead

Brain-Controlled Robotic Arm National Innovation Program, HUST Advisor: Dr. Dawei Ye

- Built a real-time **EEG brain-computer interface** controlling a robotic arm; applied CSP + SVM after signal preprocessing (noise filtering, artifact removal), reaching **82.4% accuracy** on five-DoF right-hand motor intent with sub-second latency

HONORS & AWARDS

- **Outstanding Graduate**, HUST
- **Champion**, HUST Debate Competition
- **Academic Excellence Scholarship** (×2) & **Self-Motivation and Diligence Scholarship**, HUST

06/2026

SKILLS

Languages: IELTS 7.5

Programming: Python, PyTorch, TensorFlow, C, MATLAB, Verilog, SPSS, Keil

Tools: CARLA, ROS, Docker, Linux, Git, L^AT_EX